

ANTÓNIO JOSÉ FREITAS MACHADO DÓRDIO GOMES

MECHANICAL ENGINEER — STRUCTURAL ANALYSIS & COMPOSITE MATERIALS

PROFILE

Mechanical engineer specialised in **structural analysis and composite materials**, with experience in finite element modelling, bonded joints and experimental validation. His work includes the numerical modelling of low-velocity impact damage in Glass/PPS composite laminates and the structural design of CFRP components for Formula Student electric race cars. Background spanning aeronautical structures coursework at FEUP, a Master's research thesis at Politecnico di Milano, and two seasons of Formula Student prototype vehicle development. Strong command of Abaqus-based simulation across static, fatigue, impact and non-linear structural regimes.

TECHNICAL FOCUS

FINITE ELEMENT ANALYSIS

ABAQUS

COMPOSITE STRUCTURES

IMPACT & DAMAGE MECHANICS

BONDED JOINTS

STRUCTURAL FATIGUE & FAILURE

EXPERIMENTAL VALIDATION

PYTHON / MATLAB

SOLIDWORKS · CAD

NON-LINEAR MECHANICS

CONTENTS

01
UAVision
Mechanical Design Engineer

02
Master's Thesis
Impact Damage · Glass/PPS
Politecnico di Milano

03
Formula Student
CFRP Wishbones &
Adhesive Joints

04
Formula Student
Drivetrain &
Transmission

05
Aeronautical Structures
Wing-Box Static &
Fatigue Analysis

06
Aircraft Structural
Materials & Joints
3 Sub-Projects

07
Non-Linear Mechanics
Buckling of Hollow
Al 7075-T6 Tube

08
WEG Internship
Thermal Modelling
Electric Motors

-
Certifications
Awards &
Skills

Mechanical Design Engineer

UAVision

Electronics integration, sheet-metal design and prototype development across aerial, ground and counter-UAS platforms.

PRODUCTS	Veteras · MPAV-S · UC DL Z10
DESIGN	Fusion 360 · Sheet metal · FDM
MANUFACTURING	Laser cutting · Press brake operation
KEY RESULT	23% lower total jammer assembly weight



VETERAS



MPAV-S



UC DL Z10

PRODUCT IMAGES © UAVISION
Veteras / MPAV-S / UC DL Z10

Mechanical Design Engineer

Jun 2026 - Present · Fusion 360 CAD · Sheet metal · Hands-on production · FDM

PRODUCT CONTEXT

Veteras - All-electric heavy-lift VTOL with up to 200 kg payload.

MPAV-S - Multipurpose, modular UGV for logistics and casualty evacuation.

UCDL Z10 - Modular, gyrostabilised anti-UAS jammer.

ENGINEERING CONTRIBUTIONS

- **Integrate electronics** into Veteras, MPAV-S and UCDL Z10 through mechanical component design.
- Design **sheet-metal parts in Fusion 360** and personally **operate laser cutting and press brake machines** for in-house production.
- **FDM-print parts for electronics integration**, selecting filaments to suit each application.
- **Develop prototypes and improve existing products**, primarily Veteras and UCDL Z10.

UCDL Z10 · WEIGHT REDUCTION



UCDL Z10 · OFFICIAL PRODUCT IMAGE © UAVISION

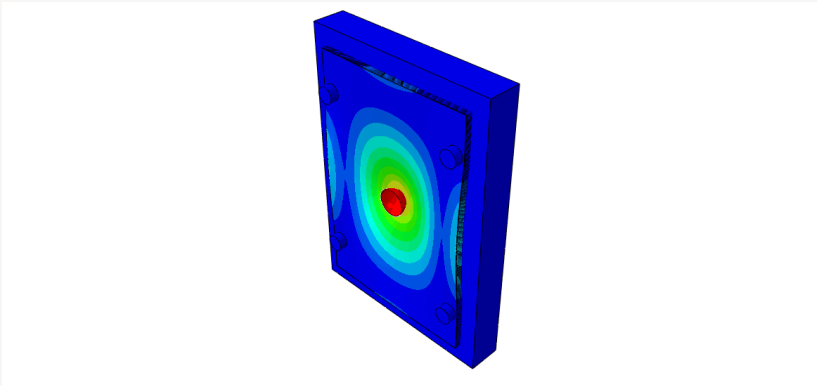
23%

REDUCTION IN TOTAL
JAMMER ASSEMBLY WEIGHT

Replaced machined components with sheet-metal parts in the **UCDL Z10**, reducing **total jammer assembly weight by 23%**. Designed and manufactured the replacement parts in-house.

Impact Damage Modelling in Glass/PPS Thermoplastic Composite Laminates

Graduate Researcher — Department of Aerospace Science and Technology
Politecnico di Milano, Italy



FINITE ELEMENT SIMULATION OF IMPACT-INDUCED DEFORMATION IN THE GLASS/PPS LAMINATE MODEL (ABAQUS/EXPLICIT)

MATERIAL	Woven Glass/PPS thermoplastic composite laminates
SOLVER	Abaqus/Explicit · Hashin + CZM damage formulation
VALIDATION	Drop-weight impact tests · Ultrasonic C-scan
KEY RESULT	Delamination area predicted within ~15% of C-scan data

Low-Velocity Impact Modelling of Glass/PPS Laminates

Graduate Researcher · Sep 2025 – Mar 2026

OBJECTIVE

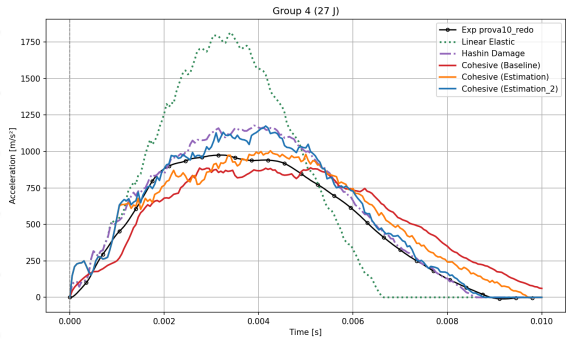
Develop and validate an Abaqus/Explicit finite element model for low-velocity impact in woven Glass/PPS thermoplastic laminates, capable of reproducing the global impact response and the main damage mechanisms observed experimentally.

METHODOLOGY

- Implemented **intralaminar damage modelling** in Abaqus/Explicit using the Hashin 2D criterion
- Introduced **cohesive interface elements** to represent interlaminar delamination
- Developed a **bi-phasic shell–solid modelling strategy** to investigate matrix-dominated deformation and permanent indentation beneath the impactor
- Reproduced the **drop-weight impact configuration** through calibrated boundary conditions, contact definitions and impact loading
- Compared numerical predictions against **experimental impact response** and **ultrasonic C-scan damage measurements**

VALIDATION APPROACH

- Compared predicted **delamination footprints** with ultrasonic **C-scan** measurements
- Correlated numerical **acceleration–time response** with experimental impact data
- Assessed the capability of different modelling strategies to reproduce **back-face cracking, delamination**, and **local indentation behaviour**



IMPACT RESPONSE CORRELATION

Acceleration–time comparison between experiment and FE models

KEY RESULTS

~15%

DELAMINATION AREA PREDICTION ERROR VS
ULTRASONIC C-SCAN

Bi-phasic

IMPROVED REPRESENTATION OF LOCAL
INDENTATION BEHAVIOUR

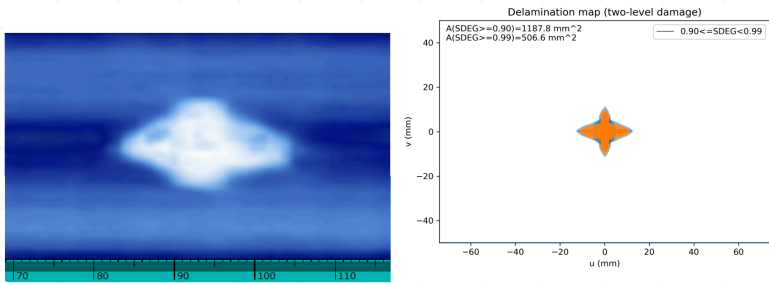
ENGINEERING CONTRIBUTIONS

- Identified the relative roles of **intralaminar fracture**, **interlaminar delamination**, and **matrix-dominated deformation** in the impact response
- Demonstrated that **intralaminar damage modelling** provided the best compromise between accuracy and computational cost for reproducing the global response and back-face crack morphology
- Showed that **cohesive modelling** was necessary to relate numerical predictions to **ultrasonic delamination measurements**
- Used the **bi-phasic approach** to investigate permanent indentation mechanisms not captured by shell-only laminate models

TOOLS & METHODS

- Abaqus/Explicit · Python post-processing
- Ultrasonic C-scan inspection · Drop-weight impact testing
- Finite element damage modelling · Experimental correlation

MODEL VALIDATION



DELAMINATION FOOTPRINT COMPARISON

Ultrasonic C-scan vs cohesive damage prediction

Electric Formula Racing Prototype Development

Composite suspension, bonded joints and transmission engineering
Formula Student FEUP



FSFEUP ELECTRIC PROTOTYPE DEVELOPED FOR THE 2025 COMPETITION SEASON

ROLES	Wishbones & Vibrations Lead · Transmission Engineer
FOCUS	CFRP structures · Adhesive bonded joints · Fatigue · Lightweight drivetrain design
COMPETITIONS	FS Germany · FS Portugal · FS United Kingdom · FS Netherlands · FS Switzerland
HIGHLIGHTS	Design, Cost & Manufacturing and Business Plan awards across 2024–2025

CFRP Suspension Wishbones with Bonded Aluminium Connectors

Wishbones & Vibrations Lead · Formula Student FEUP
Electric Formula Racing Car



FORMULA STUDENT REAR SUSPENSION SHOWING CFRP WISHBONES AND ALUMINIUM HUB ASSEMBLY

MATERIALS	CFRP tubes · 7075-T6 aluminium connectors · structural adhesive
SCOPE	25 suspension load cases · ≈40 bonded joint specimens tested
SIMULATION	Abaqus/Standard · Cohesive Zone Modelling (CZM)
KEY ACHIEVEMENT	Bi-adhesive bonding solution validated to 80°C operating temperature

CFRP Suspension Wishbones with Bonded 7075-T6 Aluminium Connectors

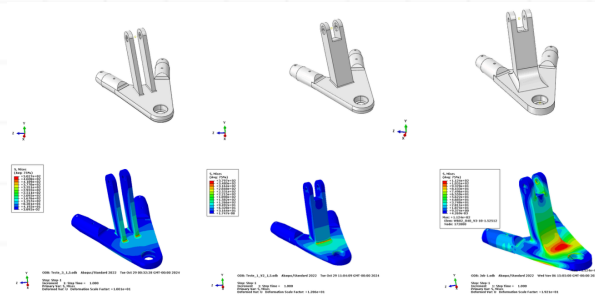
Led a 2-person sub-team · Structural simulation, bonded-joint validation, manufacturing and experimental testing

OBJECTIVE

Manufacture, assemble and validate lightweight CFRP suspension wishbones with bonded 7075-T6 aluminium end connectors for the Formula Student electric race car, ensuring structural performance across suspension load cases and experimental validation of the bonded-joint concept up to 80°C.

STRUCTURAL ANALYSIS

- Performed **Abaqus/Standard simulations of aluminium connectors** across **25 suspension load cases**
- Modelled the adhesive layer using **Cohesive Zone Modelling (CZM)** to evaluate joint failure load and damage evolution
- Correlated simulation predictions with tensile test data from **representative bonded tube–connector specimens**
- Evaluated the team's **bi-adhesive joint concept** through FEM correlation and experimental testing, focusing on **tensile strength, repeatability** and **failure mode**



FE ANALYSIS OF 7075-T6 ALUMINIUM CONNECTOR UNDER SUSPENSION LOAD CASES

MANUFACTURING & TESTING

- Manufactured and tested **≈40 representative CFRP tube–connector specimens** in tensile loading to failure
- Assembled and integrated bonded wishbone prototypes, troubleshooting bonding and manufacturing issues during implementation
- Validated the bonded-joint concept experimentally from **room temperature to 80°C**
- Compared observed fracture patterns and failure loads with **numerical predictions**

KEY RESULTS

25

SUSPENSION LOAD CASES EVALUATED IN FE SIMULATION

80°C

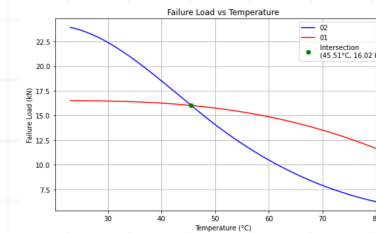
BONDED JOINT VALIDATED AT OPERATING TEMPERATURE

≈40

BONDED JOINT SPECIMENS MANUFACTURED AND TESTED

5

TEMPERATURE LEVELS TESTED FROM RT TO 80°C

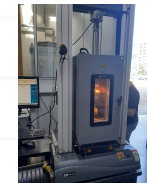


FAILURE LOAD VS TEMPERATURE COMPARISON BETWEEN JOINT SOLUTIONS

MANUFACTURING & TESTING SETUP



MANUFACTURING JIG AND ASSEMBLED WISHBONES



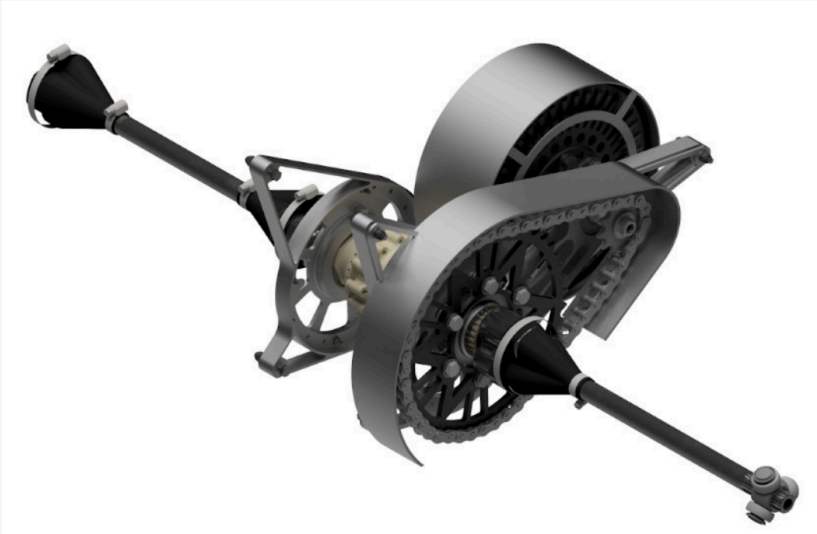
TENSILE TESTING WITH ENVIRONMENTAL CHAMBER

ENGINEERING CONTRIBUTIONS

- Continued validation of the team's **bi-adhesive wishbone joint concept** through simulation and experimental testing
- Established correlation between **cohesive FE modelling** and **experimental bonded-joint behaviour**
- Supported suspension subsystem implementation through **manufacturing, assembly** and **troubleshooting**

Drivetrain & Transmission Design

Transmission Engineer — Formula Student FEUP
Electric Formula Racing Car FSFEUP-02



FSFEUP-02 DRIVETRAIN ASSEMBLY — DIFFERENTIAL, TRIPOD HOUSING, HALFSHAFTS AND CHAIN DRIVE

COMPONENTS	Tripod housing · Driveshaft · Halfshafts · Sprocket adaptor
ANALYSIS	Static FEM · Safe-life fatigue · Topology optimisation
MASS SAVED	4 kg drivetrain reduction vs previous prototype
TRIPOD HOUSING	35% weight reduction: 1.407 kg → 0.920 kg

Drivetrain Development: Tripod Housing Optimisation, Spline Verification & Fatigue Analysis

Transmission Engineer · Structural simulation, drivetrain integration, fatigue assessment and interface validation

SCOPE

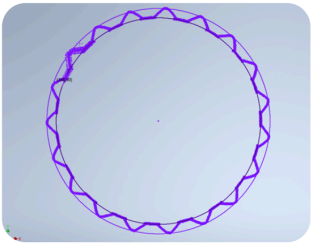
Developed and validated key drivetrain components for the FSFEUP-02 electric prototype, with emphasis on reducing subsystem mass, accommodating a new lower-cost automotive tripod joint, and ensuring structural integrity through FEM, spline verification and safe-life fatigue design.

STRUCTURAL DEVELOPMENT

- Performed **static FEM analysis** of the tripod housing and sprocket adaptor in Abaqus, assessing stresses and safety factors under drivetrain torque loads
- Iterated the **tripod housing** through three design stages, moving from a two-material concept to a fully steel optimised solution
- Validated the final tripod housing under **880 Nm torsional loading**, achieving a verified structural safety factor after mass reduction
- Executed **safe-life fatigue analysis** of the halfshafts using LTS torque spectra and S-N based assessment methods

TRIPOD JOINT & INTERFACE VALIDATION

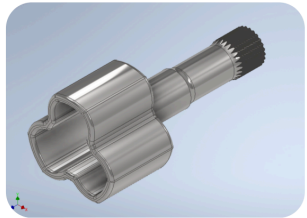
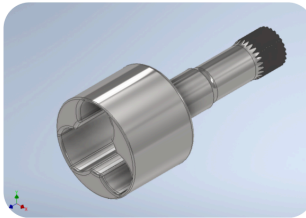
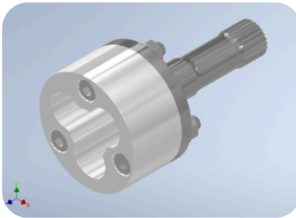
- Adopted a **street-car tripod joint** solution to **reduce cost by 800€** and simplify local manufacturability compared with custom FSAE tripod interfaces
- Performed **spline reverse engineering** using an involute projector and CAD reconstruction of the FIAT 127 tripod interface
- Verified that the new tripod interface follows **DIN 5480 N 21 × 1 × 19**, enabling a more standardised spline definition for manufacturing
- Used the **LTS torque spectrum** as input for safe-life halfshaft dimensioning and fatigue validation



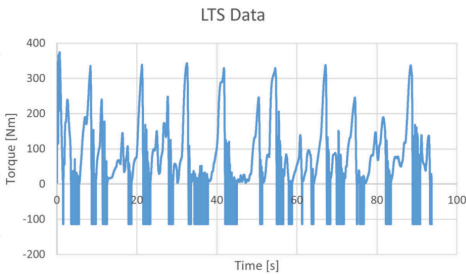
SPLINE RECONSTRUCTION AND DIN 5480 VERIFICATION OF THE FIAT 127 TRIPOD INTERFACE

WEIGHT REDUCTION RESULTS

COMPONENT	INITIAL	FINAL	REDUCTION
Tripod Housing	1.407 kg	0.920 kg	-35%
Sprocket Adaptor	0.251 kg	0.123 kg	-51%
Drivetrain Total	—	-4 kg	vs prev. prototype



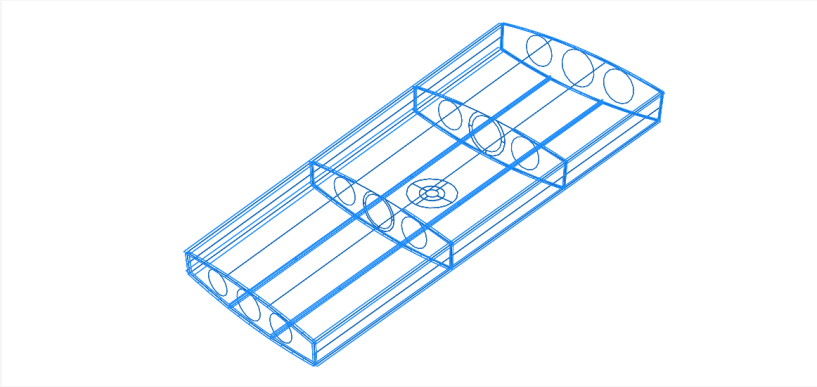
TRIPOD HOUSING EVOLUTION FROM INITIAL CONCEPT TO FINAL LIGHTWEIGHT DESIGN



TORQUE SPECTRUM FROM LAP-TIME SIMULATION USED FOR HALFSHAFT FATIGUE ASSESSMENT

Structural & Aerospace Engineering

Finite Element Analysis · Structural Mechanics · Failure Mechanics
Selected projects in aerospace structures and advanced structural analysis



FINITE ELEMENT ANALYSIS OF AN AIRCRAFT WING-BOX STRUCTURE

TOPICS	Aircraft wing-box analysis · Composite materials · Nonlinear structural behaviour
METHODS	Finite Element Analysis · Buckling · Fatigue analysis · Failure criteria
TOOLS	Abaqus · MATLAB · Python · Analytical structural methods
PROJECTS	Wing-box structural analysis · Aircraft materials & joints · Nonlinear mechanics

Aircraft Wing-Box Static Structural Analysis

Abaqus FEM · MATLAB · Excel · Aerospace structural verification

OBJECTIVE

Evaluate the structural behaviour of an aircraft wing-box under limit load conditions and determine certification safety margins for the main structural members including skins, spars and ribs.

METHODOLOGY

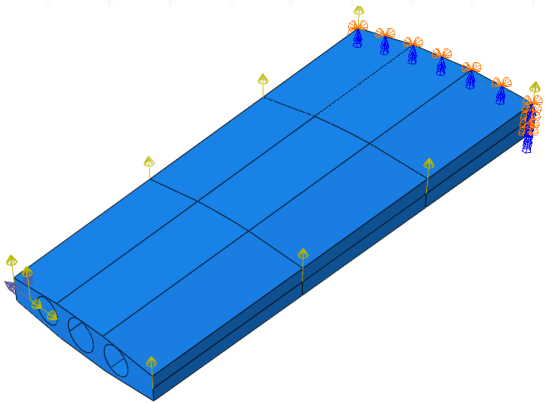
- Developed a detailed **Abaqus finite element model** of the wing-box including skins, spars, ribs and stringers
- Applied representative **aerodynamic loads and boundary conditions** to simulate wing structural response
- Extracted stress fields, internal forces and displacement distributions from FEM results
- Post-processed results in **MATLAB and Excel** to compute margins of safety for multiple failure modes
- Evaluated structural certification criteria including **tension, global buckling, local buckling and crippling**

KEY RESULTS

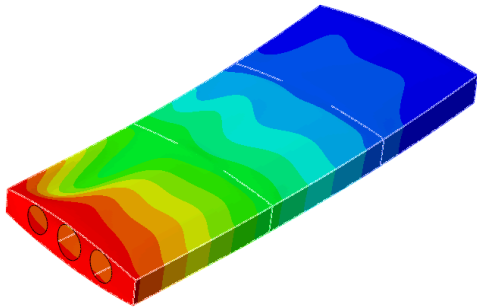
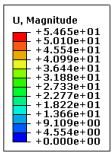
COMPONENT	FAILURE MODE	MS
Spar S1	Global Buckling	-0.60
Spar S2	Global Buckling	-0.59
Skin P1	Normal Crippling	-0.31
Rib N2	Tension	10.52

ENGINEERING INSIGHT

- Negative safety margins in **spars S1 and S2** indicate susceptibility to global buckling under compressive loading
- Skin panel P1** shows insufficient normal crippling resistance, suggesting the need for structural reinforcement or geometry modification
- The analysis highlights the importance of **buckling-driven design constraints** in lightweight aerospace wing structures



FINITE ELEMENT MODEL OF THE WING-BOX WITH APPLIED LOADS AND BOUNDARY CONDITIONS



GLOBAL DISPLACEMENT FIELD OF THE WING-BOX UNDER LIMIT LOAD CONDITIONS

Aircraft Wing-Box Fatigue & Damage Tolerance Analysis

Composite failure criteria · Safe-life fatigue · Crack growth modelling

OBJECTIVE

Assess the fatigue behaviour and damage tolerance of an aircraft wing-box, combining composite ply-level verification of the upper skin with safe-life and inspection-based evaluation of critical metallic components.

METHODOLOGY

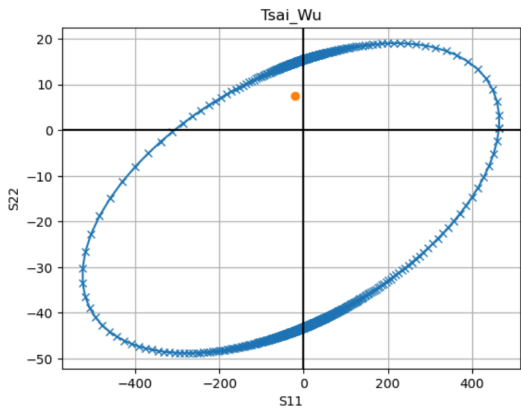
- Modified the wing-box model to include a **[0/+45/-45/90]_s CFRP upper skin laminate** and local stress-raising geometric features
- Evaluated composite ply behaviour under two gust load cases using **Maximum Stress, Maximum Strain, Tsai-Hill and Tsai-Wu** criteria
- Computed **stress concentration factors** for the inspection hole and rib N3 central hole
- Derived **fatigue notch factors** using Neuber-based relations
- Performed **safe-life fatigue analysis** for Rib N3 and **damage tolerance analysis** for the lower inspection hole using crack-growth methods

KEY RESULTS

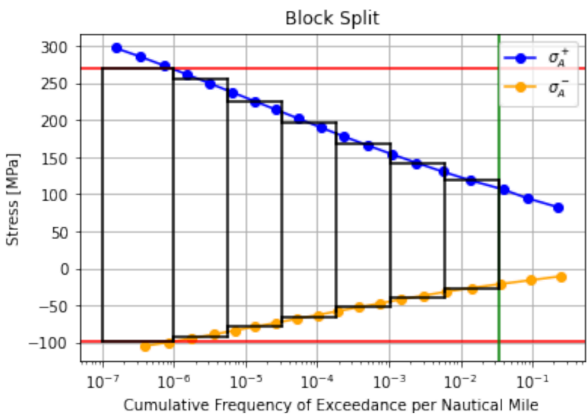
COMPONENT	ASSESSMENT	RESULT
Upper Skin (CFRP)	Infinite Life	Plies 4 & 5 not certified
Rib N3	Safe Life	7.38×10^5 FH
Inspection Hole	First Inspection	2.30×10^7 FH
Inspection Hole	Re-inspection	3,026 FH

ENGINEERING INSIGHT

- The **CFRP upper skin satisfies static certification**, but the two 90° plies fall below the required infinite-life fatigue margin under the critical gust case
- Rib N3 achieves safe-life certification**, demonstrating adequate fatigue resistance under the derived mission spectrum
- The inspection-hole study shows how **damage tolerance methods** can be used to define inspection planning rather than immediate redesign



TSAI-WU FAILURE ENVELOPE USED FOR COMPOSITE PLY STRENGTH ASSESSMENT



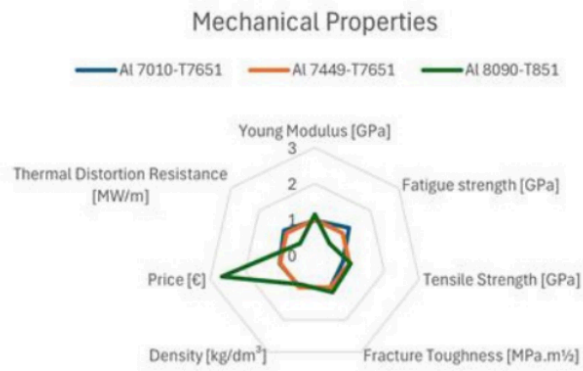
STRESS SPECTRUM AND BLOCK DISCRETIZATION USED FOR FATIGUE LIFE ASSESSMENT

Aircraft Structural Materials – Light Alloys & Composite Laminate Mechanics

Material selection · MATLAB · Classical Laminated Plate Theory · Monte Carlo analysis

1 – LIGHT ALLOYS: AIRBUS A380 WING RIB MATERIAL SELECTION

- Investigated **premature crack formation in Airbus A380 wing rib feet**, associated with the use of Al 7449 and its lower fatigue and thermal distortion resistance for this application
- Performed a **material selection study** using literature review and the **GRANTA EduPack Level 3 Aerospace database** to compare candidate aluminium alloys for wing rib structures
- Analysed the trade-off between **Al 7449 and Al 7010** in terms of fatigue strength, fracture toughness, stiffness and thermal distortion resistance
- Extended the study to alternative alloys from the **2000 and 8000 series**, including Al 2024-T351, Al 2090-T83 and Al 8090-T851
- Engineering conclusion:** Al 7010 provides the most balanced solution for rib feet, while Al 8090 offers attractive specific stiffness but insufficient fatigue performance for safe-life design

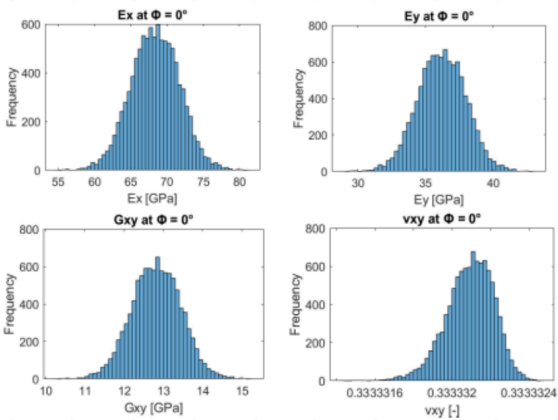


SELECTED MECHANICAL PROPERTIES COMPARISON

ALLOY	Σ_{UTS} [MPa]	FATIGUE [MPa]	K_{IC} [MPa√m]
Al 7449-T7651	605	219	29.6
Al 7010-T7651	488	269	29.6
Al 2024-T351	392	140	36.4
Al 8090-T851	504	142	34.0

2 – COMPOSITE LAMINATE MECHANICS

- Developed **MATLAB routines** to determine the effective elastic properties of AS4/3501-6 CFRP laminates from **micro-mechanics**, using Rule of Mixtures and **Tsai's modulus / invariant approach**
- Applied **Classical Laminated Plate Theory (CLPT)** to multidirectional laminates and **Double-Double laminates $[\pm\Phi/\pm\Psi]$** , evaluating the evolution of E_x , E_y , G_{xy} and ν_{xy} with ply orientation
- Performed **Monte Carlo sensitivity analyses** to quantify the effect of constituent Young's modulus variability, ply-angle deviations of $\pm 5^\circ$, and their combined influence on laminate stiffness
- Compared predicted laminate behaviour across different Φ/Ψ angle combinations, showing the design flexibility of **Double-Double laminate architectures**
- Engineering insight:** even small manufacturing deviations in ply angles can propagate into measurable variability in laminate engineering constants, reinforcing the importance of **uncertainty quantification** in composite design



3 — Adhesive Bonding of Aluminium–Composite Structures

Experimental testing · Abaqus cohesive modelling · Failure prediction

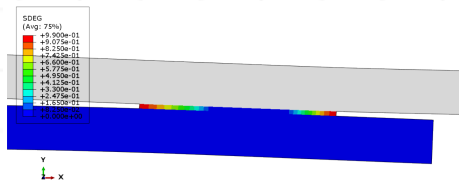
ADHESIVE JOINT INVESTIGATION

- Manufactured and tested **aluminium–CFRP single-lap joints (SLJ)** to investigate bonded-joint structural behaviour.
- Compared two aerospace structural adhesives: **Araldite AV138** (epoxy paste adhesive) and **AF 163-2K** (film adhesive commonly used in aerospace structures).
- Specimens were prepared with controlled adhesive thickness and **25 mm overlap length**, following bonded-joint manufacturing procedures.
- Performed **tensile testing** using a universal testing machine to determine failure load and failure mode.

FINITE ELEMENT MODELLING

- Developed a **2D Abaqus model** to simulate bonded-joint mechanical behaviour.
- Adherends were discretized with **CPE4R elements** and the adhesive layer with **COH2D4 cohesive elements**.
- The cohesive-zone model captured **damage initiation and propagation** in the adhesive layer.
- Simulation results were compared with experiments to **validate failure prediction capability**.

ARALDITE AV138 — ADHESIVE FAILURE



FEM PREDICTION

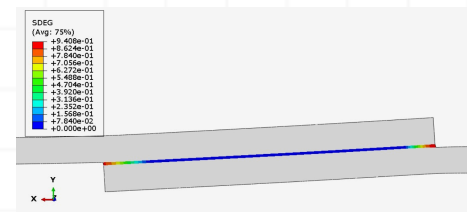


EXPERIMENTAL FAILURE

EXPERIMENTAL VS NUMERICAL RESULTS

- **Araldite AV138**: failure load ≈ 5.16 kN, with cohesive failure in the adhesive layer.
- **AF 163-2K**: failure load ≈ 18.5 kN, with failure occurring in the **aluminium adherend**.
- Finite element predictions showed good agreement with experiments ($\approx 4\%$ error for AV138 and $\approx 7.6\%$ error for AF163-2K).
- Results confirmed the superior structural performance of **AF 163-2K film adhesive** for aluminium–composite bonding.

AF 163-2K — ADHEREND FAILURE



FEM PREDICTION



EXPERIMENTAL FAILURE

Non-Linear Buckling of a Hollow Al 7075-T6 Tube

Abaqus/Standard · Linear Perturbation · Static, Riks · Imperfection Sensitivity

OBJECTIVE

Investigate the buckling behaviour of a thin-walled aluminium tube and compare linear eigenvalue and non-linear FEM predictions, including the effects of geometric imperfections and material plasticity.

MODEL DEFINITION

PARAMETER	VALUE
Geometry	Ø 28 mm · t = 2 mm · L = 1000 mm
Material	Al 7075-T6
Element	S8R shell
BCs	Bottom fixed · top pinned + axial load

ANALYSIS WORKFLOW

- Validated FEM against **Euler analytical buckling load**
- Extracted first **buckling eigenmode** using linear perturbation
- Introduced **geometric imperfection** from eigenmode before the Riks step
- Performed **Static, Riks analysis** with and without plasticity

KEY RESULTS

0.67%

ERROR BETWEEN EULER
ANALYTICAL SOLUTION AND
FEM EIGENVALUE RESULT

181 kN

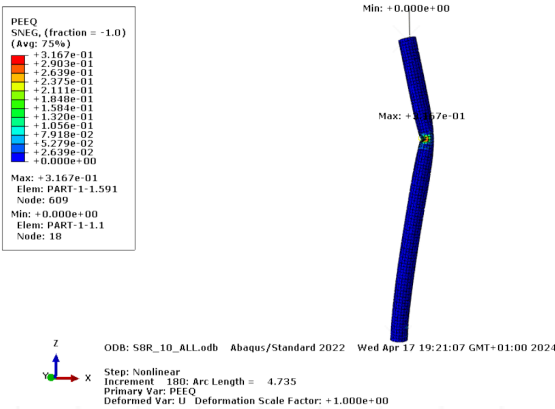
CRITICAL BUCKLING LOAD
FROM LINEAR PERTURBATION
ANALYSIS

159 kN

FULLY NON-LINEAR
BUCKLING LOAD INCLUDING
PLASTICITY

ENGINEERING INSIGHT

- Imperfection implementation** is essential to activate realistic buckling behaviour in Riks analysis
- Including both **geometric and material non-linearity** reduced the predicted buckling load relative to the idealized model
- Plastic strain localised at the buckle region, demonstrating interaction between **instability and yielding**



PLASTIC STRAIN LOCALISATION IN THE BUCKLE REGION DURING FULLY NON-LINEAR RIKS ANALYSIS

Thermal Modelling of Heat Dissipation in Electric Motor Housings

Product Development Department · Bachelor's Thesis Internship · ANSYS Steady-State Thermal

OBJECTIVE

Evaluate the heat dissipation performance of electric motor housings during the transition from cast-iron housings to steel housings with welded fins. Develop and validate ANSYS thermal FEM models to support design decisions regarding fin geometry and joining configuration.

METHODOLOGY

- Developed **steady-state thermal FEM models in ANSYS** to simulate heat dissipation in motor housings
- Compared **cast-iron housings** with **welded steel fin configurations**
- Performed **parametric analysis of fin geometry**: chamfered vs non-chamfered welded fins
- Conducted **experimental validation** using PT100 sensors, controlled airflow and thermographic imaging
- Used **Python scripts** for heat-transfer calculations and result post-processing

KEY RESULTS

34–40%

HEAT-DISSIPATION LOSS OF CHAMFERED STEEL FINS RELATIVE TO CAST-IRON HOUSING

48–52%

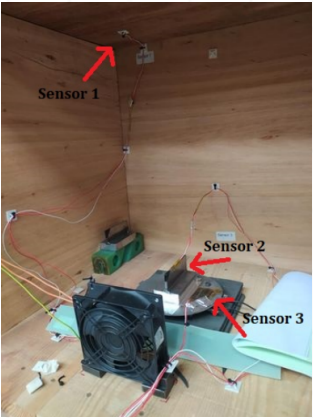
HEAT-DISSIPATION LOSS OF NON-CHAMFERED STEEL FINS RELATIVE TO CAST-IRON HOUSING

7.7%

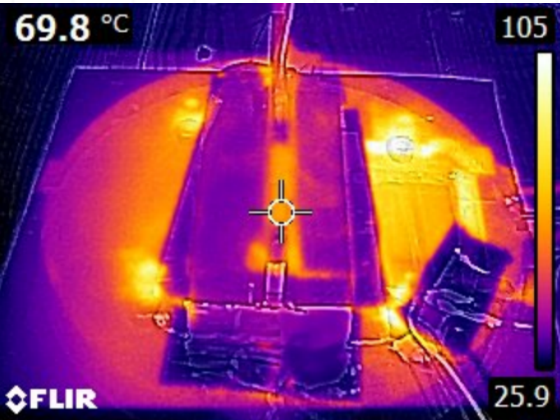
MAXIMUM DEVIATION BETWEEN CORRECTED FEM MODEL AND EXPERIMENTAL MEASUREMENTS

ENGINEERING INSIGHTS

- Chamfered steel fins improved thermal contact and clearly outperformed non-chamfered fins
- Thermal contact resistance and radiation effects were essential to match simulation and experiment
- The study combined **thermal simulation, test setup design and experimental validation** in an industrial context



EXPERIMENTAL SETUP WITH PT100 SENSORS AND FORCED AIRFLOW FOR THERMAL VALIDATION



THERMOGRAPHIC MEASUREMENT OF HOUSING TEMPERATURE DISTRIBUTION DURING TESTING

TOOLS USED

- ANSYS Workbench · Python · SolidWorks · Excel · Thermal instrumentation

FEA & SIMULATION

- **Abaqus** — Explicit, Standard, Riks; damage, CZM, topology optimisation
- **ANSYS** — Thermal FEM
- **Python** — Scripting, parametric model automation
- **MATLAB** — CLPT, homogenisation, fatigue, data processing

CAD & DESIGN

- **SolidWorks** — CSWA Certified (Aug 2023)
- **Inventor · Fusion 360**
- **KISSoft** — Gear and spline analysis

DOCUMENTATION

- LaTeX · Excel · PowerPoint

LANGUAGES

- **Portuguese** — Native
- **English** — C1
- **Spanish** — B1
- **French** — A2

CERTIFICATIONS

SW

CSWA — Certified SolidWorks Associate
Dassault Systèmes · Aug 2023
Credential: C-2JAUPXEG85

ST

Composites Design Workshop XXVII
Prof. Steve Tsai — Stanford University
Aeronautics & Astronautics · Jul 2024

ENGINEERING FOCUS AREAS

COMPOSITE STRUCTURES & DAMAGE MECHANICS

FINITE ELEMENT ANALYSIS (ABAQUS/EXPLICIT & STANDARD)

STRUCTURAL FATIGUE & DAMAGE TOLERANCE

IMPACT DAMAGE MECHANICS

ADHESIVE BONDED JOINTS & CZM

NON-LINEAR STRUCTURAL MECHANICS

AERONAUTICAL STRUCTURAL DESIGN

EXPERIMENTAL VALIDATION METHODS

FORMULA STUDENT AWARDS 2025 (EV CLASS)

1st

BUSINESS PLAN
FS PORTUGAL

2nd

ENGINEERING DESIGN
FS PORTUGAL

3rd

COST REPORT
FS PORTUGAL

3rd

COST REPORT
FS GERMANY

9th

BUSINESS PLAN
FS GERMANY

1st

BUSINESS PLAN
FS NETHERLANDS

FORMULA STUDENT AWARDS 2024 (CONCEPT CLASS)

1st

OVERALL
FS SWITZERLAND

1st

DESIGN & COST
FS SWITZERLAND

2nd

OVERALL
FS UK

1st

COST REPORT
FS UK

1st

OVERALL
FS PORTUGAL

1st

DESIGN & COST
FS PORTUGAL

ENGINEERING PORTFOLIO · 2026

ANTÓNIO GOMES

MECHANICAL ENGINEER
AERONAUTICAL STRUCTURES & COMPOSITES

AREAS OF INTEREST

- Aerospace Structures
- Composite Materials & Structural Simulation
- Motorsport & High-Performance Engineering

Open to graduate engineering roles in structural simulation, composite materials and aerospace structures.

Thank you for your time and consideration.

CONTACT INFORMATION

LOCATION	Porto, Portugal
EMAIL	antmachado7@gmail.com
PHONE	+351 939 290 065
LINKEDIN	linkedin.com/in/antmachado7

EDUCATION

MSc in Mechanical Engineering –
Aeronautical Structures & Vehicles
FEUP · 2023 - 2026 · Thesis at Politecnico di
Milano

BSc in Mechanical Engineering
FEUP · 2020 – 2023